DS-2002: Data Systems

An Overview of the SQL Language

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The SQL Language

Understanding the Structured Query Language

The SQL Language: Principal Components

Three Primary Aspects of the ANSI-Compliant SQL Language

Data Definition Language (DDL)

- CREATE, ALTER, DROP, TRUNCATE TABLE, ENABLE & DISABLE TRIGGER
- *Used to manage database structures*

Data Control Language (DCL)

- GRANT, REVOKE, DENY, EXECUTE AS
- *Used to control access to server & database objects (permissions)*

Data Manipulation Language (DML)

- SELECT, INSERT, UPDATE, DELETE, MERGE, and BULK INSERT
- *Used to manipulate database content (data)*

Microsoft Docs | Transact-SQL Reference (Database Engine)

Query a SQL Database: The SELECT Statement

Essential Components of Data Retrieval

SELECT	for specifying the required columns
FROM	for specifying the Table(s) being targeted
JOIN	for specifying additional Table(s)
GROUP BY	for specifying points of aggregation
ORDER BY	for sorting rows of the result set

Filtering Statements:

ON	WHERE	HAVING
specifies the column(s) that enable the joining of two Tables	specifies conditions by which to reduce the rows returned	specifies conditions by which Groups or Aggregates may be reduced

Database Transactions: Enforcing Data Integrity

The **ACID** Properties of a Transaction Must be Enforced to Ensure the Integrity of a Database

Atomicity

- All operations in a transaction must be treated as a single [atomic] unit, and must either be fully committed (i.e., succeed) or be rolled-back (i.e., have no effect whatsoever).

Consistency

- Each transaction must guarantee the valid state of the database; both before and after the transaction is either committed or rolled-back.
- All defined rules, constraints, and relationships must be enforced and maintained.

Isolation

- Concurrent transactions must be completely independent; having no effect on each other.
- Transactions must remain invisible to each other until they are either committed or rolled-back.

Durability

- Once committed, the effects of each transaction must be permanently saved (written to disk); even if the system fails amidst their execution.
- A transaction log (aka, Journal file) serves as a write-ahead mechanism to ensure each committed transaction can be replayed to successfully recover the database to a consistent state in the event of a system failure.

Q & A

An Overview of SQL Databases